

# AI Mock Interview Project Documentation

## Tech stack

### Backend

- **LiveKit Agents** (core “agent runtime” + real-time interaction layer)
- **LangGraph** (complementary workflow/state machine for deterministic stage control)
- **FastAPI (Python)** or **Node/Express** (API + optional WS for UI updates)
  - If you use WebSocket in the frontend, hardcode:  
`ws://127.0.0.1:8000/ws`

### Frontend

- **React.js + CSS** (simple dashboard UI to make stage logic + fallbacks visible)
- Optional: lightweight auth gate (dummy username/password)

### Storage (for “persistent memory”)

- **In-memory** for the demo (fastest/lowest risk), keyed by `session_id`
- Optional upgrade: **Postgres** keyed by `session_id` + `stage`  
(Pinecone not needed unless you’re doing semantic search/RAG)

## Brief workflow logic

### Stages (finite state machine)

- **INTRO** → **EXPERIENCE** → **DONE**

### What runs where

- **LiveKit**: handles user interaction (voice/text), streaming responses, session events.
- **LangGraph**: enforces deterministic stage transitions + “no conflict” rules.

### Transition rules (hard-coded in logic, not prompt-only)

Each stage transitions when **any** of the following is true **and** the agent is not currently speaking/streaming:

1. **Completion criteria met** (e.g., intro collected)
2. **Turn limit reached** (`turn_count >= max_turns`)
3. **Time-based fallback** (`elapsed_stage_time >= stage_timeout`)
4. **Idle fallback** (`time_since_last_user_activity >= idle_timeout`) → nudge first; if still idle and stage timeout reached, transition.

### Anti-conflict guarantees

- Only one stage active at a time.
- Block transitions while `is_agent_speaking == true` (prevents interruptions/overlap).
- Debounce transitions so they can only fire once per stage.

## Scenarios & Edge Cases:

### 1) User takes too long to complete a turn

- Track `last_user_activity_time`
- If `idle_timeout` exceeded: supportive nudge
- If `stage_timeout` exceeded: **force stage transition** (time-based fallback)

### 2) User goes off-topic

- Detect off-topic (simple heuristic or LLM boolean classifier)
- First time: redirect + restate question
- Repeated off-topic: count strikes; still obey turn/time limits to avoid loops
- If overall stage exceeds time: transition regardless

### 3) Persistent memory without leakage

You'll implement **two memory layers**, both sandboxed and sanitized:

**Outer memory: Session-level (very basic)**

- Keyed by `session_id`
- Stores only minimal session metadata + *sanitized* transcript summary (not raw)
- Purpose: prevent cross-session bleed, enable basic observability

**Inner memory: Stage-level (INTRO/EXPERIENCE sandboxes)**

- Keyed by `(session_id, stage)`
- Each stage stores only its own state:
  - `turn_count`, timers, transition reasons, stage-specific summary
- **No raw transcript is shared across stages**

- Only a **sanitized handoff summary** is passed from INTRO → EXPERIENCE

## Sanitization/anonymity rules (applies before persisting anything)

### Data minimization

- Default: do **not** persist raw transcripts
- Persist only:
  - allowlisted fields (e.g., role, top skills, general background)
  - sanitized summaries
  - metrics (turns, durations, transition reason)

### PII redaction (best-effort)

- Emails → [REDACTED\_EMAIL]
- Phone numbers → [REDACTED\_PHONE]
- Addresses/IDs → [REDACTED]
- Optionally: keep only first name or redact names entirely

### Stage handoff

- INTRO produces **handoff\_summary\_safe**
- EXPERIENCE receives only **handoff\_summary\_safe** + allowlisted fields
- EXPERIENCE never reads INTRO raw content

## Demo UI features (React)

Core panels:

- **Stage badge** (INTRO / EXPERIENCE / DONE)
- **Transcript viewer** (what user sees; can be non-persisted)
- **Timers**
  - elapsed stage time

- idle time
  - “next fallback” (time remaining until stage\_timeout)
  - **Diagnostics**
    - turn count (e.g., 2/3)
    - last transition reason (criteria\_met | timeout | turn\_limit | idle\_timeout)
    - agent speaking status (true/false)
  - **Controls:**
    - Start / Stop session
    - Optional debug toggle (force transition button hidden behind toggle)
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